

Day 19 - Lecture 19.1: Histograms in Matplotlib

1. Overview & Purpose

A **Histogram** is a graphical representation of the distribution of numerical data. It groups continuous data into discrete ranges called **bins** and counts how many values fall into each bin (frequency).

Why use a Histogram?

- **Understand Data Distribution:** Quickly observe whether data is symmetric (normal), skewed (left/right), bimodal, or uniform.
 - **Identify Spread & Central Tendency:** See where the concentration of data lies (mean, median, mode region).
 - **Detect Outliers & Gaps:** Identify unusual spikes or regions where no data exists.
 - **Histogram vs. Bar Chart:**
 - **Bar Chart:** Used for categorical/discrete categories (e.g., fruits, cities, movie titles). Bars have gaps between them.
 - **Histogram:** Used for continuous quantitative variables (e.g., marks, age, height, salary). Bars typically touch each other because intervals are continuous.
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2. Core Concepts & Parameters

```
flowchart TD
    A["Raw Continuous Data"] --> B["Divide Range into Bins/Intervals"]
    B --> C["Count Frequencies per Bin"]
    C --> D["Plot Bars: Width = Bin Range, Height = Frequency"]
```

`plt.hist()` Function Syntax:

```
plt.hist(
    x,                # 1D array-like data
    bins=None,        # Integer (number of equal bins) or sequence (custom
    range=None,        # (lower, upper) range of bins
    density=False,     # If True, returns probability density (area sums to
    cumulative=False,  # If True, computes cumulative histogram
    color='skyblue',   # Fill color of the bars
    edgecolor='black', # Outline color of bars (crucial for distinguishing a
    linewidth=1.2,     # Border thickness
```

```
alpha=1.0,          # Transparency level (0.0 transparent to 1.0 opaque)
orientation='vertical' # 'vertical' (default) or 'horizontal'
)
```

Binning Strategies:

1. **Integer Bins:** `bins=10` splits $[\min(x), \max(x)]$ into 10 equally spaced intervals.
2. **Custom Bins (Sequence):** `bins=[0, 30, 70, 90, 100]` creates unequal bins: $[0, 30)$, $[30, 70)$, $[70, 90)$, $[90, 100]$. This is ideal for grade boundaries (e.g., Fail, Average, Good, Excellent).

3. Mathematical & Statistical Foundation

1. Bin Width Calculation:

$$\Delta = \frac{\max(x) - \min(x)}{k}$$

where k is the number of bins.

2. Common Rules for Optimal Number of Bins (k):

- **Sturges' Rule** (Best for normal data):

$$k = 1 + \lceil \log_2(n) \rceil$$

- **Freedman-Diaconis Rule** (Resistant to outliers):

$$\text{Bin Width } h = 2 \cdot \frac{\text{IQR}(x)}{\sqrt[3]{n}}, \quad k = \left\lceil \frac{\max(x) - \min(x)}{h} \right\rceil$$

4. Code Implementation

```
import matplotlib.pyplot as plt
import numpy as np
```

```
# Set random seed for reproducibility
np.random.seed(0)

# Generate 100 exam scores with mean = 70 and standard deviation = 10
scores = np.random.normal(loc=70, scale=10, size=100)

# 1. Basic Histogram
plt.figure(figsize=(8, 5))
plt.hist(scores, bins=10, color='skyblue', edgecolor='black', alpha=0.8)
plt.title("Distribution of Exam Scores (Equal Bins)")
plt.xlabel("Marks / Scores")
plt.ylabel("Number of Students (Frequency)")
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()

# 2. Custom Bins (Grading System)
marks = [
    25, 38, 42, 55, 61, 67, 72, 78, 83, 88,
    91, 95, 47, 53, 59, 64, 69, 74, 79, 85,
    92, 97, 33, 45, 58, 62, 71, 76, 81, 89
]
grade_bins = [0, 33, 60, 80, 100]

plt.figure(figsize=(8, 5))
plt.hist(marks, bins=grade_bins, color='lightgreen', edgecolor='black', linewidth=2)
plt.title("Grade Distribution of 30 Students (Custom Bins)")
plt.xlabel("Score Intervals")
plt.ylabel("Student Count")
plt.xticks(grade_bins)
plt.grid(True, linestyle=':', alpha=0.6)
plt.show()
```

5. Key Takeaways & Interview Points

1. **Always specify `edgecolor`** : Without `edgecolor='black'` , adjacent bars blend together into a single monochromatic block.
2. **Too many vs. Too few bins:**
 - *Too few bins (underfitting)*: Oversimplifies the distribution, hiding modes and clusters.
 - *Too many bins (overfitting)*: Creates a jagged, noisy plot dominated by sampling noise.
3. **Horizontal orientation:** Use `orientation='horizontal'` when comparing against vertical axes or side-by-side marginal distributions.